

Architectural Pattern Selection

Incident Escalation & On-Call Rotation Engine

Architectural Choice

We chose Microservices Architecture for the Incident Escalation & On-Call Rotation Engine. The system is divided into independent components for alert ingestion, incident management, on-call rotation, notification and escalation, and post-mortem management. This architecture allows each major function to operate as a separate service while communicating through defined interfaces.

Justification

The first reason for selecting Microservices Architecture is the separation of responsibilities in the system. Incident alert handling, on-call rotation management, notification and escalation, and post-mortem management are distinct functional areas. Separating them into independent components makes the system easier to maintain, modify, and extend without affecting unrelated functionality.

The second reason is the system's time-sensitive incident handling requirement. Alerts must trigger notification through SMS, email, or webhook within 3 seconds of incident ingestion. The Notification & Escalation component can be independently scaled and optimized to handle time-critical notification and escalation operations, while other services continue performing their respective functions.

Security Implications

Microservices support role-based access control by allowing access to sensitive operations to be restricted according to user roles. For example, only authorized users should be able to create or modify on-call schedules and incident post-mortems. Separating these functions into dedicated services also provides clearer boundaries for enforcing authorization and protecting sensitive incident information.

Performance Implications

The architecture supports improved performance by allowing services involved in critical operations to be independently scaled. In particular, the Notification & Escalation component can be optimized for rapid dispatch of incident notifications, helping the system satisfy the requirement that alert notifications are initiated within 3 seconds of incident ingestion.

Conclusion

Therefore, Microservices Architecture is suitable for the Incident Escalation & On-Call Rotation Engine because it provides modularity, independent scalability, clear security boundaries, and efficient handling of time-sensitive incident workflows.